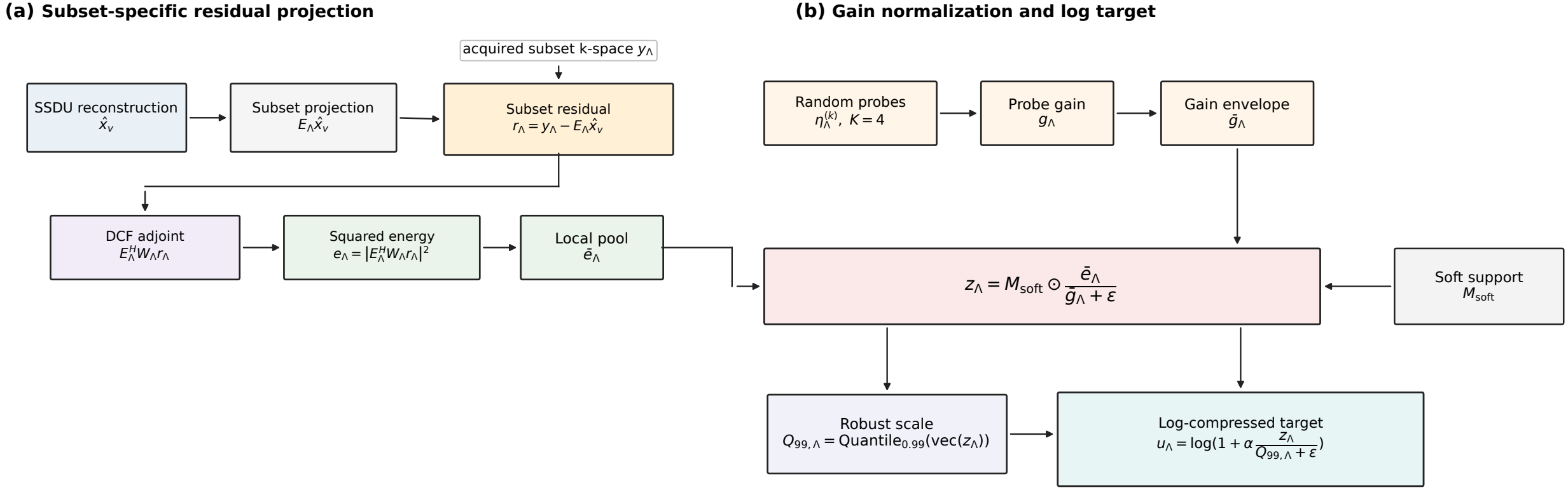
(a) Subset-specific residual projection
acquired subset k-space y_Λ
SSDU reconstruction
$\hat{x}_v$
Subset projection
$E_\Lambda \hat{x}_v$
Subset residual
$r_\Lambda = y_\Lambda - E_\Lambda \hat{x}_v$
DCF adjoint
$E_\Lambda^H W_\Lambda r_\Lambda$
Squared energy
$e_\Lambda = |E_\Lambda^H W_\Lambda r_\Lambda|^2$
Local pool
$\bar{e}_\Lambda$
(b) Gain normalization and log target
Random probes
$\eta_\Lambda^{(k)},\ K = 4$
Probe gain
g_Λ
Gain envelope
$\bar{g}_\Lambda$
$z_\Lambda = M_{\text{soft}} \odot \frac{\bar{e}_\Lambda}{\bar{g}_\Lambda + \varepsilon}$
Soft support
M_{soft}
Robust scale
$Q_{99,\Lambda} = \text{Quantile}_{0.99}(\text{vec}(z_\Lambda))$
Log-compressed target
$u_\Lambda = \log(1 + \alpha \frac{z_\Lambda}{Q_{99,\Lambda} + \varepsilon})$